

The AI talent war: Geography of AI invention and US-China inventor mobility

Artificial intelligence (AI) is increasingly central to economic growth and national competitiveness, placing a premium on access to elite AI talent. Using comprehensive pre-grant patent application data and the OECD’s AI classification method, we examine the geography of AI innovation and the cross-border mobility of AI inventors, with a focus on the United States and China. We document a growing mismatch between the locations of AI-inventing firms and AI inventors: although US firms continue to dominate AI patenting, the share of AI inventors residing in the United States has stagnated, while China’s AI activity has expanded rapidly. Tracking inventor mobility reveals an increasingly asymmetric pattern, with experienced and highly productive AI inventors more likely to move from the United States to China than in the opposite direction. Given the substantial concentration of AI innovation among top inventors, these talent flows may have outsized implications for technological leadership. Our findings highlight the importance of strategic human capital policies—particularly those governing immigration and talent retention—that strengthen long-term AI talent capacity without inadvertently accelerating talent outflows.

1 INTRODUCTION

The growth of artificial intelligence (AI) has already had, and will continue to have, a profound impact on nearly every dimension of the global economy. As AI technologies increasingly shape productivity, innovation, and economic growth, both firms and governments view leadership in AI as critical to long-term competitiveness and national security.

Central to this competition is the scarcity of elite AI talent. Developing top-tier AI researchers and engineers requires years of advanced training and accumulated experience, and the global supply of such talent remains extremely limited. For example, Meta has reportedly engaged in aggressive hiring efforts, offering compensation packages exceeding \$100 million to recruit leading AI engineers (Nolan, 2025). Moreover, cutting-edge AI development depends on individuals who combine deep technical expertise with the ability to integrate knowledge across multiple disciplines. The rarity of individuals with such breadth and depth has made AI talent a key bottleneck in technological progress.

As a result, the global “war for talent” has become a policy concern as much as a corporate one. Governments increasingly recognize that access to and retention of top AI innovators directly shape national technological leadership. This dynamic is especially salient for the United States and China, which have long been at the forefront of AI research and development and now compete most intensely for scarce AI talent.

2 WHERE ARE AI INVENTIONS PRODUCED, AND HOW HAS THIS CHANGED?

We examine the geographic distribution of AI invention and its evolution over time, with particular attention to the cross-border mobility of AI inventors between the United States and China, the world's two leading hubs for AI research and development. In this context, the growing outflow of highly productive and experienced AI inventors may generate persistent direct and indirect consequences for technological leadership in AI.

We use patent application data from PatentsView, which provide standardized, time-stamped, and globally comparable information on inventive activity and inventor locations. We focus on pre-grant patent applications, which offer the most timely and accurate measure of inventive activity and inventor mobility during the period of intensifying U.S.–China competition in AI. In contrast, granted patents typically lag by two to three years, and many AI-related inventions remain pending. Note that alternative data sources such as self-reported LinkedIn profiles are not best suited for this context because LinkedIn penetration in China is limited due to access restrictions, and individuals who relocate from the United States to China often stop maintaining their profiles.

To identify which of the 5.3 million U.S. patent applications published from 2011 through the third quarter of 2025 involve AI, we apply the latest semi-automated method suggested by the OECD (OECD, 2025). This approach that builds on earlier methodologies and are validated through expert reviews and interviews classifies AI inventions using a combination of patent's Cooperative Patent Classification (CPC) subclasses and AI-related keyword filters. Using this protocol, 4.25% of all applications during 2011–2025Q3 are categorized as core AI (1.94%) and AI-related (2.31%) patents.

2.1 The Geography of AI Invention and Its Evolution

We first document and illustrate the global trends. **Fig. 1** shows the top 10 countries ranked by firms with largest AI patents and how it evolved in the past fifteen years, shown separately by the location of firms (panel A) and the location of inventors (panel B).

Our first finding is that the geographic distribution of AI-inventing firms does not necessarily align with the distribution of AI inventors. The United States continues to host the largest—and increasingly growing—share of AI-inventing firms. However, the share of U.S.-based AI inventors has remained largely stagnant in recent years, suggesting that U.S. firms increasingly rely on foreign AI talent. In contrast, China exhibits close alignment between the locations of AI-inventing firms and inventors, with both shares rising over time.

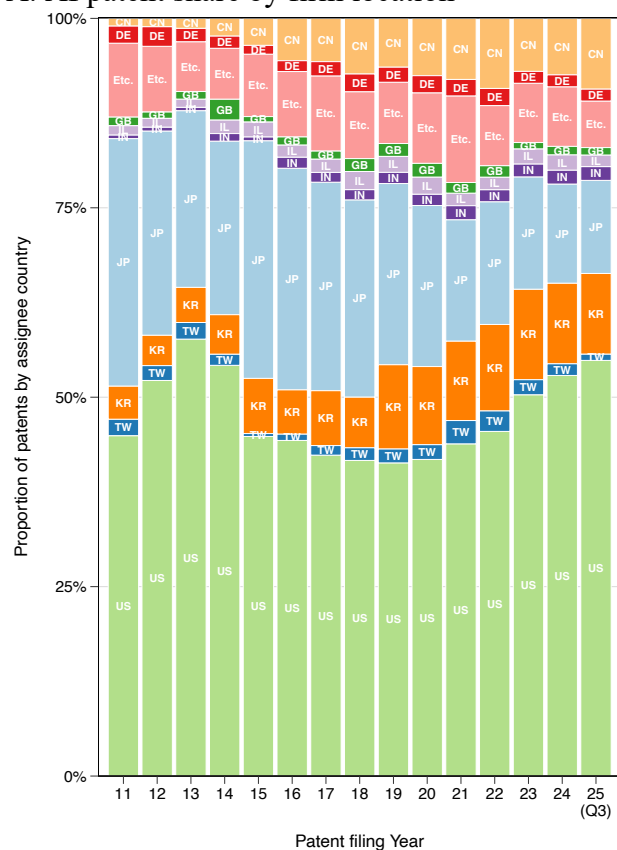
Beyond the United States and China, inventors based in Canada and India account for a substantial share of AI patenting, despite relatively limited AI-inventing activity by firms located in these

countries. This pattern suggests that these countries generate highly capable AI inventors, while domestic firms may have limited capacity to absorb or retain such talent—resulting in outward or remote forms of AI talent migration (brain drain).

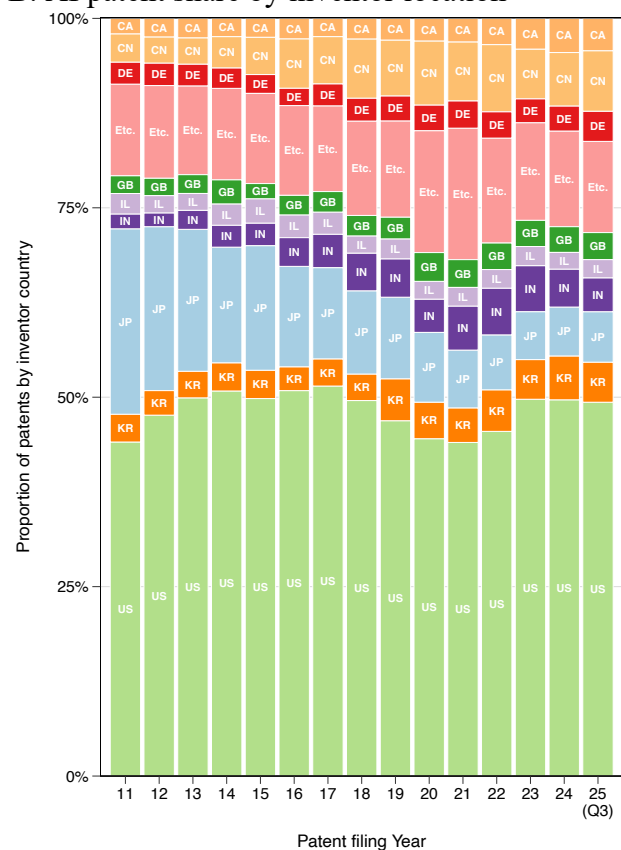
Taiwan, along with Korea and Japan, presents the opposite pattern. Taiwanese firms contribute disproportionately to global AI patenting, yet the share of inventors located in Taiwan remains comparatively small and falls outside the top ten globally. This disparity indicates that Taiwanese firms rely heavily on foreign inventors or geographically distributed R&D operations.

Fig. 1. Country-level AI patent shares. (A). AI patent share by firm location. (B). AI patent share by inventor location. Results are shown for the top nine countries, with all other countries aggregated as *Etc.* (C). AI patent application counts for the United States and China. (D). AI patent applications normalized by each country's average during 2011–2015.

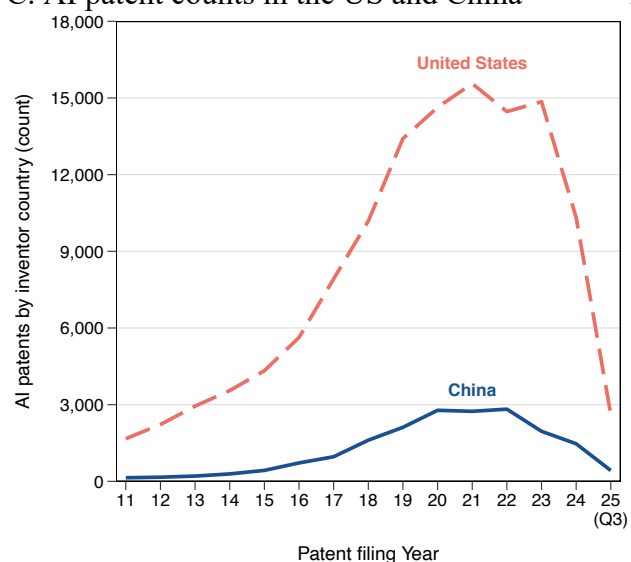
A. AI patent share by firm location



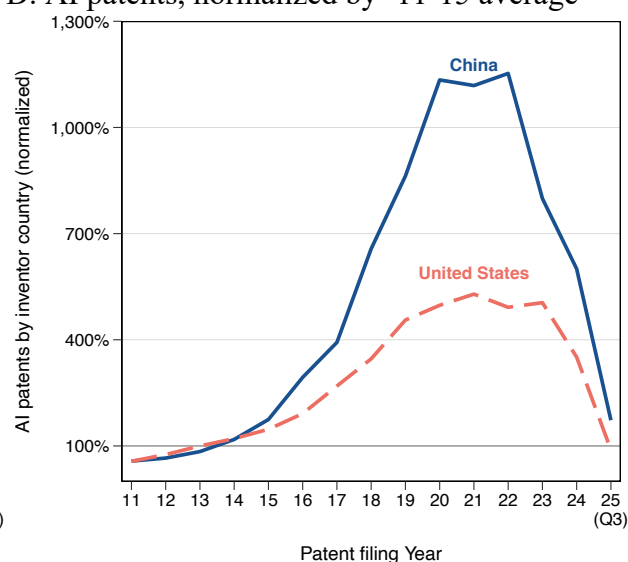
B. AI patent share by inventor location



C. AI patent counts in the US and China



D. AI patents, normalized by '11-15 average



Firms in the United States and China together account for 60.9% of all AI patent applications in 2024. A similar pattern emerges when examining inventor residence: inventors residing in these two countries produced 66.3% of all AI-related inventions in 2024. **Fig. 1**, panels C and D, illustrates how the production of AI inventions has evolved over time in both countries. Note that the decline observed in the most recent years is mechanical; a portion of recently filed patents has not yet appeared in the pre-grant application data, and for 2025 we only observe filings from the first three quarters of the year, resulting in right-tail truncation. This truncation, however, does not compromise our analysis as long as there is no systematic reason to expect that publication lags or truncation occur differentially across the two countries.

Fig. 1C shows that the total number of AI patent applications produced by inventors residing in the United States consistently exceeds those produced by inventors residing in China. It is important to note that this classification is based on inventors' locations at the time of filing, not their nationality—meaning that Chinese-origin inventors working in the United States are coded as U.S.-based inventors. At first glance, this pattern that US-based inventors dominate AI patent filings may appear inconsistent with growing concerns that the United States is losing its AI leadership to China.

However, **Fig. 1D**, which normalizes patent counts by each country's 2011–2015 average, offers a more nuanced picture. When relative growth is considered, the rate of increase in China's AI patenting activity far surpasses that of the United States. Thus, while the United States currently maintains dominance in terms of the accumulated volume of AI inventions, China's much faster growth trajectory suggests that this leadership may erode over time if current trends persist.

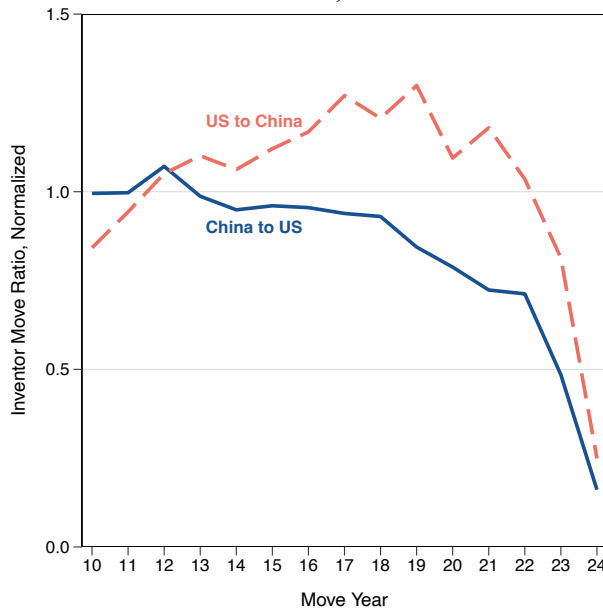
2.2 AI Inventor's Mobility Between the United States and China

From these AI patents, we identify inventors who have contributed to at least one AI patent filing and classify them as AI inventors. We then track their geographic mobility over time. Specifically, if an inventor appears on a patent filed in year t with a location different from their immediately preceding patent, we infer that the inventor moved to the new location in the prior year ($t-1$). This allows us to systematically construct a detailed picture on year-to-year cross-border mobility of AI talent.

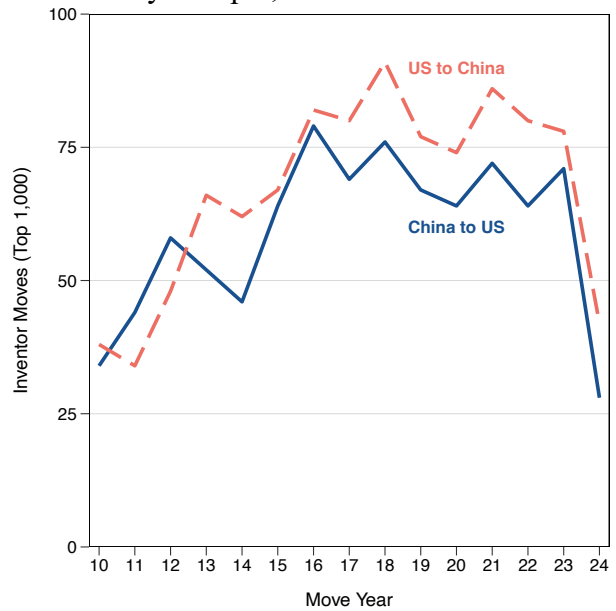
Fig. 2A shows cross-border move ratio of AI inventors (i.e., outgoing movers as a share of all AI inventors in each country) between the United States and China, normalized by each category's 2011–2015 average for easier comparison. Mobility from China to the United States falls below 100% and exhibits a declining trend, whereas mobility from the United States to China exceeds 100% and has increased over time, widening the gap between the two flows. This divergence has become more pronounced in recent years (except for the most recent two years, which are affected by right-tail truncation) indicating a growing net outflow (brain drain) of AI inventors from the United States.

Fig. 2. Mobility of AI Inventors between the United States and China. (A). AI inventor move ratio (i.e., movers as a share of all AI inventors) between the United States and China, normalized by each flow's average during 2011–2015. (B). Mobility of the top 1,000 AI inventors between the United States and China. (C). Mobility of AI inventors by experience (“rookies” for bottom quartile and “veterans” for top quartile), normalized by the 2011–2015 average. (D) Stayers (i.e., non-movers) who remain in a single country throughout their observed careers, normalized by the 2011–2015 average.

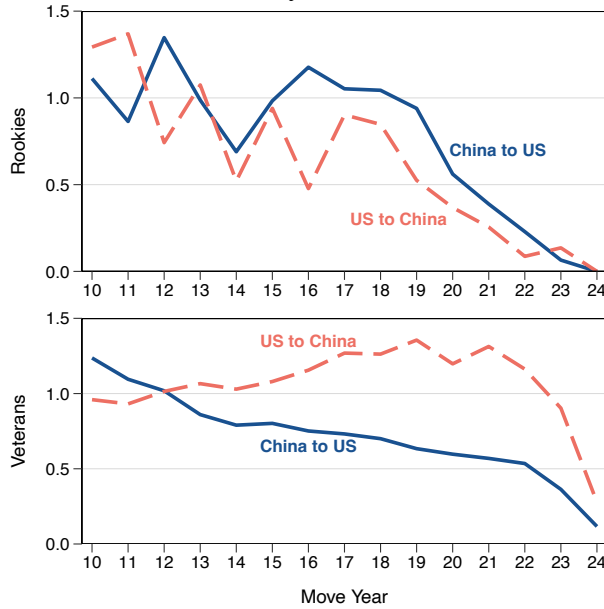
A. AI inventor move ratio, normalized



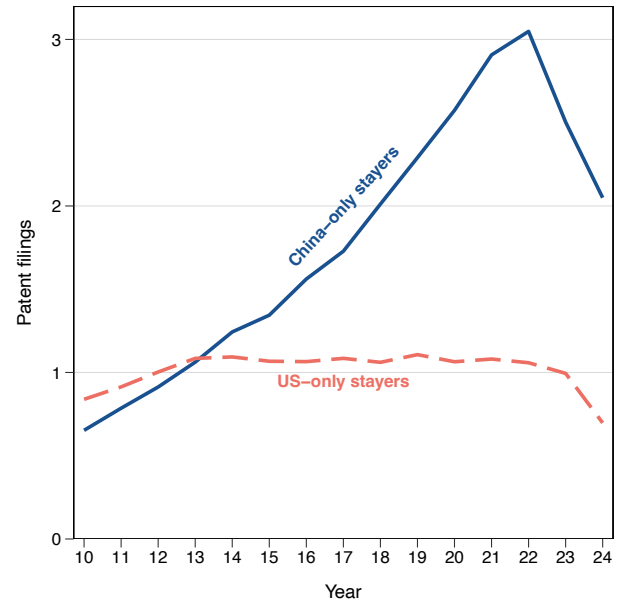
B. Mobility of top 1,000 AI inventors



C. AI inventor mobility: Rookie vs. veterans



D. Stayers in a single country (non-mobility)



To better understand these trends, we examine the composition of inventor mobility. A distinctive feature of the AI field is the high concentration of inventive output among a small number of inventors; over 2022–2025, the average CR100 (the patent share of the top 100 inventors) was 5.3% in AI, compared with 2.5% in non-AI fields. This concentration underscores the importance of analyzing the characteristics of mobile inventors beyond aggregate mobility flows.

Fig. 2B shows mobility patterns among the top 1,000 AI inventors ranked by lifetime patent productivity. Since 2016, movements of top AI inventors from the United States to China have exceeded flows in the opposite direction, indicating a net outflow of top AI talent from the United States. **Fig. 2C** reveals a similar pattern for highly experienced AI inventors (those in the top quartile by years since first patenting). In contrast, rookie inventors seeking training, exposure, or early-career opportunities are more likely to move to the United States. Taken together, these patterns indicate that the most productive and experienced AI inventors are increasingly more likely to move from the United States to China than vice versa.

To the extent that these senior inventors have accumulated experience, tacit knowledge, professional networks, and other valuable assets during their time in the United States, their relocation may facilitate the transfer of such assets to Chinese firms, with potential implications for U.S. technological leadership. These inventors are often positioned to lead advanced AI research and development efforts. In contrast, inventors seeking training, exposure, or early-career opportunities continue to move to the United States. In this sense, the United States may increasingly attract early-stage AI talent while losing inventors as they accumulate experience and expertise and become more impactful contributors to cutting-edge AI innovation.

2.3 What Geography of AI Inventions and Talent Flows Tell Us

Our analysis paints a nuanced picture of AI innovation and talent flows between the United States and China. Although the number of China-based AI innovators—measured by assignee location and inventor location—has grown rapidly in recent years, the United States continues to lead across both dimensions.

At the same time, a growing mismatch between the location of AI inventors and the location of AI-inventing firms highlights the increasing global fragmentation of AI talent and production (Fig. 1). US-based firms produce a larger share of AI inventions than the share of AI inventors residing in the United States, and this gap has widened over time. Persistent outflows of AI inventors from the United States to other countries, notably China, may therefore carry long-term consequences for US innovative capacity.

A defining feature of AI innovation is its extreme concentration among a small number of top inventors: AI patents are nearly twice as concentrated as patents in non-AI fields. This concentration implies that the mobility of a relatively small number of highly productive inventors can have disproportionate effects on aggregate AI innovation. Consistent with this concern, recent mobility patterns reveal a pronounced asymmetry in inventor flows between the United States and China. Although the aggregate volume of US-to-China moves remains modest relative to the size of the U.S. AI workforce, inventors who do relocate tend to be more experienced and more productive than those moving in the opposite direction.

Given the central role of tacit knowledge, team-based production, and collaborative networks in AI research, the relocation of senior inventors may disrupt existing knowledge networks and depress future innovative output. Consistent with this mechanism, prior work shows that NIH-initiated investigations into failures to disclose foreign (particularly Chinese) research support led to significant productivity declines among US-based scientists collaborating with Chinese peers (Jia et al., 2024). These findings imply that the growing outflow of experienced and highly capable AI inventors from the United States to China may impose even larger productivity losses on those who remain within the U.S. innovation system.

Taken together, these results indicate that leadership in AI depends not only on domestic innovation capacity but also on the ability to attract, retain, and integrate globally scarce talent—especially at the upper tail of the productivity distribution. However, current evidence on the geographic gravity of AI innovation and the selective mobility of top inventors raises concerns about the sustainability of U.S. leadership in AI.

3 IMPLICATIONS FOR AI SCIENCE AND POLICY

The observed patterns in the geography of AI innovation and AI talent mobility carry important implications for firms, policymakers, and national innovation systems. The United States continues to attract a steady inflow of early-career Chinese AI inventors, drawn by its universities, research infrastructure, and leading technology firms. This inflow underscores the enduring strength of the US ecosystem as a training ground for frontier AI research.

At the same time, longer-term mobility trends raise strategic concerns. A disproportionate share of experienced and highly productive AI inventors eventually relocates from the United States to China. These individuals possess deep technical expertise, accumulated tacit knowledge, and leadership capacity—attributes that are especially valuable for advancing frontier AI research and commercialization. The emerging pattern suggests an imbalance in which the United States excels at nurturing early-stage talent but increasingly loses inventors once they reach peak productivity.

These findings underscore the need for a more calibrated strategic human capital policy—including immigration policy—that strengthens the AI talent base without inadvertently reducing retention or productivity. First, immigration systems that facilitate short-term entry but limit long-term residency or career progression may unintentionally accelerate the departure of capable and experienced AI talent. Strengthening pathways to permanent residency, reducing uncertainty around visa status, and offering clearer long-term prospects for high-performing AI researchers could help mitigate this loss.

Second, recent U.S. policy approaches—including trade restrictions, export controls, and heightened scrutiny of researchers with ties to China—appear poorly suited to preserving technological leadership. Such measures may increase uncertainty and erode trust within research institutions, rather than slowing talent—and thus knowledge—outflows. Consistent with this concern, our findings show that while early-career inflows to the United States remain strong, the country is increasingly losing experienced and highly productive AI inventors.

Third, firms play a critical role alongside policymakers. Redesigning career ladders, compensation structures, and research environments to offer sustained opportunities for senior AI professionals may reduce incentives to seek leadership roles abroad.

Overall, the findings point to the need for a dual-stage AI talent strategy: sustaining the United States' comparative advantage in early-career training while simultaneously improving its ability to retain mature, high-impact inventors. Without such an approach, the United States risks reinforcing a cycle in which it educates global AI talent but sees its most valuable contributors ultimately drive innovation elsewhere, with lasting consequences for national competitiveness in AI.

4 REFERENCES

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